Penn State University
College of Engineering

CMPEN 270
Digital Design

Space Station Escape Pod Launch Controller

Design Project 3

Spring 2026

Project Overview

This project implements an 8-state escape pod launch controller for a fictional research outpost, Orbis Station. The controller is a sequential digital circuit that manages the normal launch sequence and responds to emergency conditions by either aborting and returning to the initial state or accelerating the sequence depending on the current state. The design uses three flip-flops to encode eight states and combinational logic to generate the next state and output signals, and is implemented and tested using standard logic gates and flip-flops on a breadboard.

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State	Q2Q1Q0	Phase Name	Normal Description (E = 0)	Emergency Behavior (E = 1)
S0	000	**Standby**	System ready. No output. Advance to pre-check.	Hold in S0. RED on. Emergency condition is active. Do not proceed.
S1	001	**System Pre-check**	Self-tests running. GREEN on. Advance to pressurization.	Abort. Return to S0. RED on.
S2	010	**Pressurization**	Pod pressurizing. No output. Advance to fuel load.	Abort. Return to S0. RED on.
S3	011	**Fuel Load**	Fueling complete. GREEN on. Advance to systems arm.	Proceed to Systems Armed S4. RED on.
S4	100	**Systems Armed**	Pod armed. GREEN on (stand clear). Advance to launch window.	Accelerate and Jump to the intermediary stage S7. GREEN on.
S5	101	**Launch Window**	Window open. GREEN on. Advance to countdown.	Window open. Accelerate and jump to intermediary stage S7. GREEN on.
S6	110	**Countdown**	Counting down. GREEN on (stand clear). Advance to release.	Emergency GO. GREEN on. Advance to pod release S7.
S7	111	**Pod Released/Intermediary Stage**	Launched. GREEN on. Reset to S0.	Launched. GREEN on. Reset to S3.

Inputs and Outputs

- **Input** E = Emergency condition (1-bit input). E = 0 means normal operation; E = 1 means an emergency condition is active.
- **State LEDs** Q2, Q1, Q0 = Current state (outputs of the flip-flops). These should be connected to 3 LEDs to show the binary count.
- **Output G** = Green LED. Indicates a safe condition for the current state when E=0 and accelerated launch when E=1.
- **Output R** = Red LED. Indicates an abort condition for the current state when E=1.

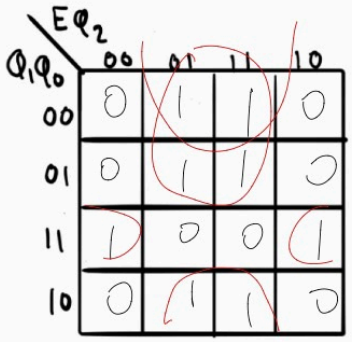
Important: G and R must **never** be ON simultaneously. Some states have both OFF (transition phases with no output).

When E=0

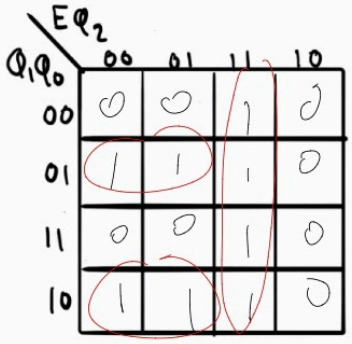
Q_2	Q_1	Q_0	Q_2^*	Q_1^*	Q_0^*	G	R
0	0	0	0	0	1	0	0
0	0	1	0	1	0	1	0
0	1	0	0	1	1	0	0
0	1	1	1	0	0	1	0
1	0	0	1	0	1	1	0
1	0	1	1	1	0	1	0
1	1	0	1	1	1	1	0
1	1	1	0	0	0	1	0

When E=1

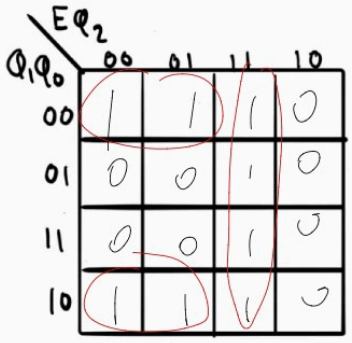
Q_2	Q_1	Q_0	Q_2^*	Q_1^*	Q_0^*	G	R
0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	1
0	1	0	0	0	0	0	1
0	1	1	1	0	0	0	1
1	0	0	1	1	1	1	0
1	0	1	1	1	1	1	0
1	1	0	1	1	1	1	0
1	1	1	0	1	1	1	0



$Q_2^* = Q_2\bar{Q}_1 + \bar{Q}_2 Q_1 Q_0 + Q_2\bar{Q}_0$

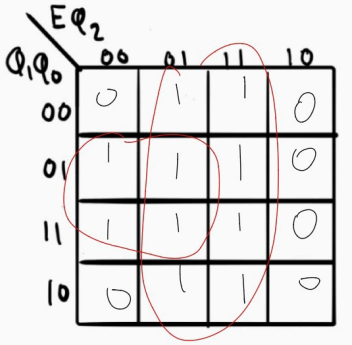


$Q_1^* = EQ_2 + \bar{E}\bar{Q}_1 Q_0 + \bar{E} Q_1 \bar{Q}_0$

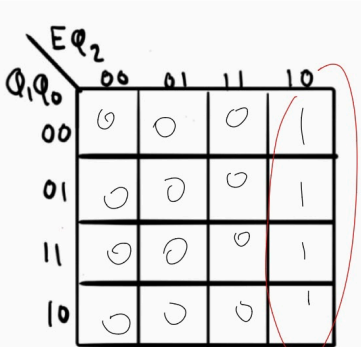


$Q_0^* = EQ_2 + \bar{E}\bar{Q}_1\bar{Q}_0 + \bar{E} Q_1 \bar{Q}_0$

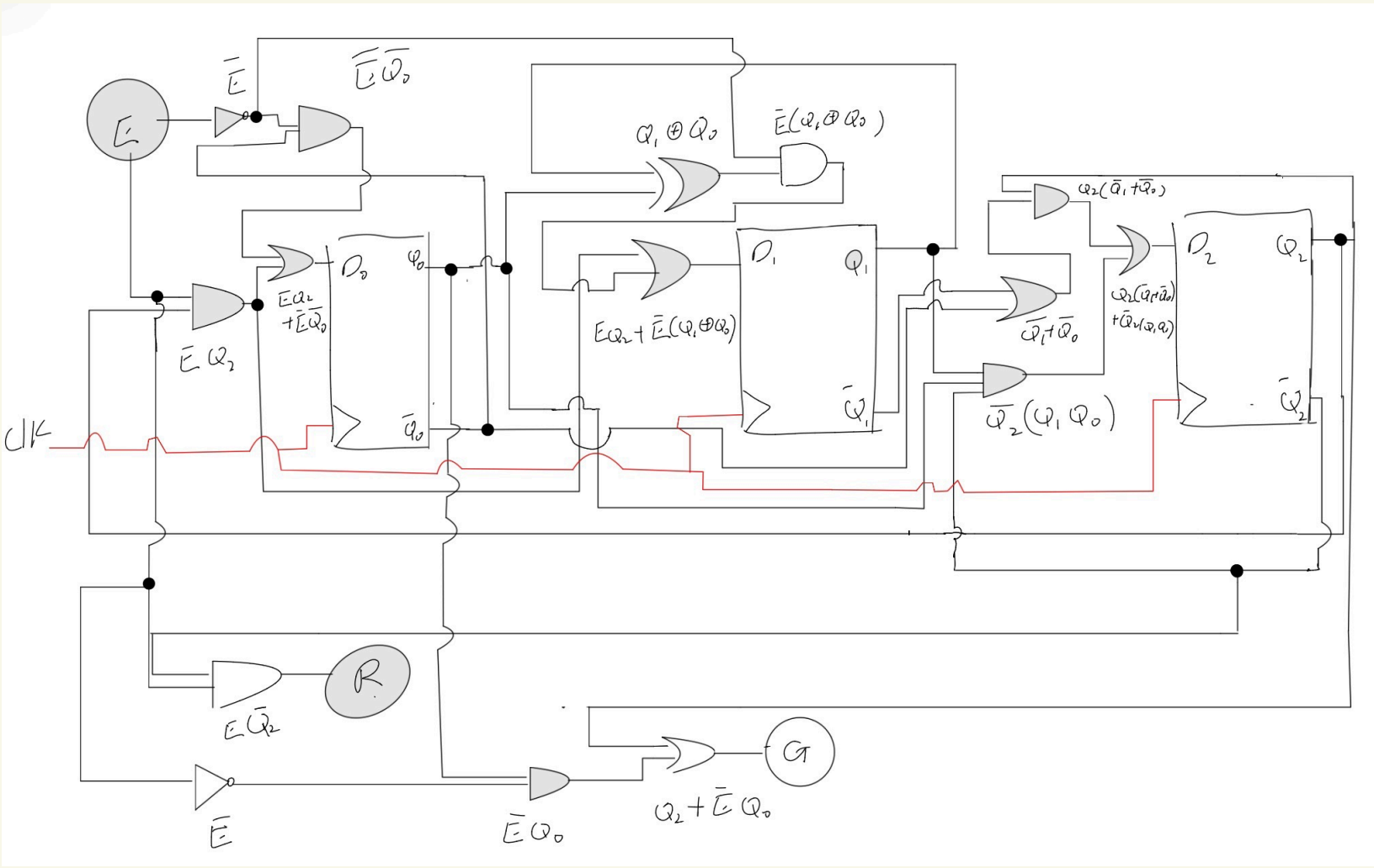
$\bar{E}(\bar{Q}_0(Q_1 + \bar{Q}_1))$

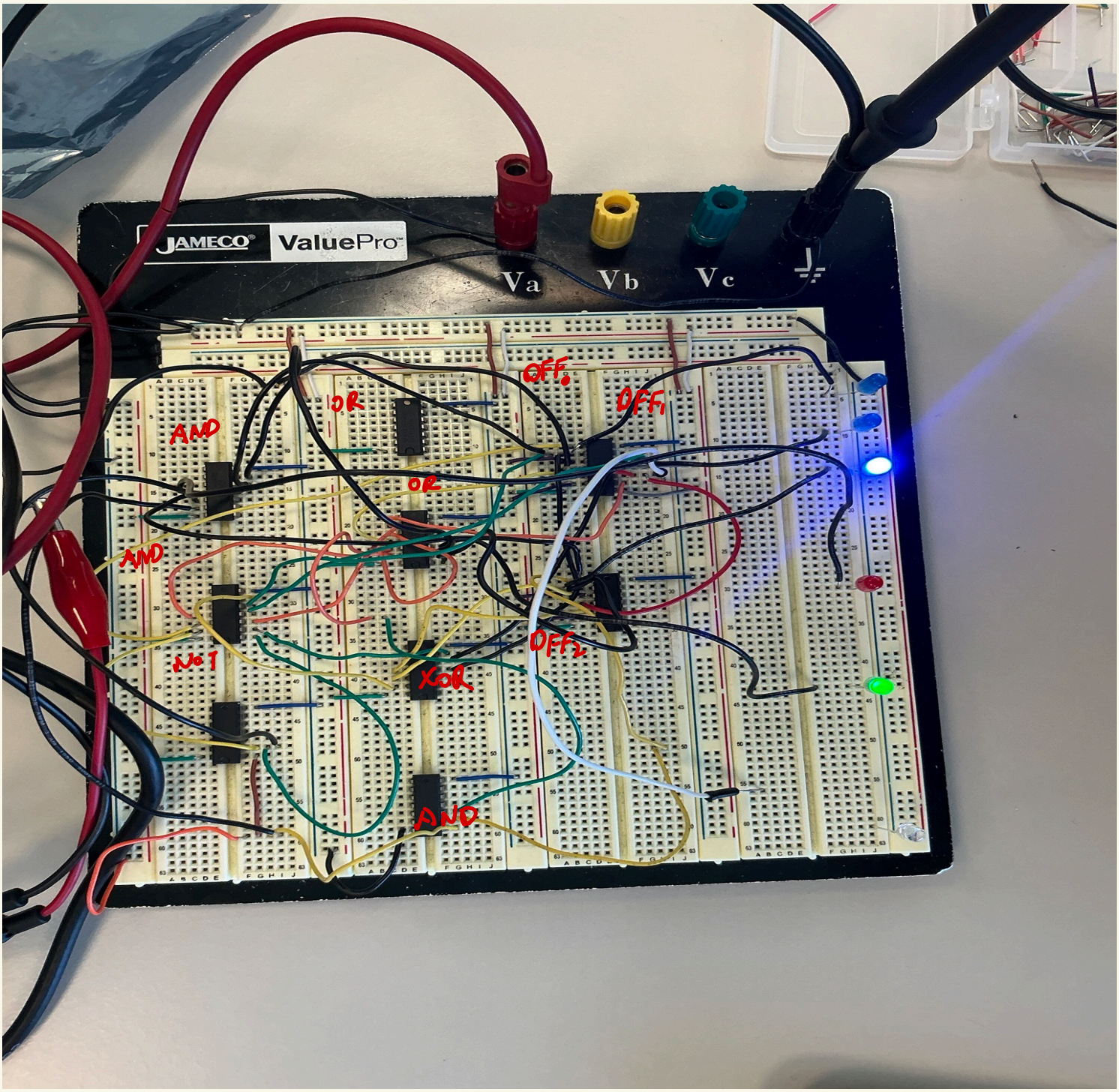


$G = Q_2 + \bar{E} Q_0$



$R = E\bar{Q}_2$


E
$\bar{E}$
$\bar{E}\bar{Q}_0$
$Q_1 \oplus Q_0$
$\bar{E}(Q_1 \oplus Q_0)$
$Q_2(\bar{Q}_1+\bar{Q}_0)$
D_0
Q_0
$\bar{Q}_0$
$E Q_2 + \bar{E}\bar{Q}_0$
$\bar{E} Q_2$
D_1
Q_1
$\bar{Q}_1$
$E Q_2 + \bar{E}(Q_1 \oplus Q_0)$
$\bar{Q}_1 + \bar{Q}_0$
D_2
Q_2
$\bar{Q}_2$
$Q_2(\bar{Q}_1+\bar{Q}_0) + \bar{Q}_2(Q_1 Q_0)$
$\bar{Q}_2(Q_1 Q_0)$
CLK
R
$E\bar{Q}_2$
G
$\bar{E}$
$\bar{E} Q_0$
$Q_2 + \bar{E} Q_0$


JAMECO ValuePro
Va
Vb
Vc
OR
OFF0
OFF1
AND
OR
AND
NOT
XOR
OFF2
AND